Algebraic Number Theory NMAL431
Exercise Sheet 2 — Solutions

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1. Let $f(x) = x^3 - 3x + 4$.

   (a) One can easily check that f has a unique root in $\mathbb{F}_7$. Computing $f(\alpha)$ for each $\alpha = 0, 1, \ldots, 6$, we get $f(\alpha) = 4, 2, 6, 1, 0, 2, 6 \pmod 7$ respectively. Thus $x = 4$ is the unique root in $\mathbb{F}_7$. We compute $f'(X) = 3X^2 - 3$, and thus $f'(4) = 45 \equiv 3 \bmod 7$.

   Therefore $v_7(f(4)) > 2v_7(f'(4))$, and thus we can apply Hensel's lemma. Thus there exists a unique root a of $f(X)$ in $\mathbb{Z}_7$, such that $a \equiv \alpha \pmod 7$. As α was the only root of $f(x) \bmod 7$, this implies $f(x)$ has a unique solution in $\mathbb{Z}_7$.

   (b) *Case $p = 3$:* Computing $f(\alpha)$ for each $\alpha = 0, 1, 2$, we get $f(\alpha) = 1, 2, 0 \pmod 3$ respectively. However, as $f'(X) \equiv 0 \pmod 3$, we cannot immediately conclude that there exists a root by Hensel's lemma. We furthermore check $f(\alpha) \bmod 9$ for each of $\alpha = 2, 5, 8$, and we get $f(\alpha) \equiv 6 \pmod 9$ in all cases. Thus f has no roots in $\mathbb{Z}/9\mathbb{Z}$, and so has no roots in $\mathbb{Z}_3$.

   Case $p = 5$: Computing $f(\alpha)$ for each $\alpha = 0, 1, 2, 3, 4$, we get $f(\alpha) = 4, 2, 1, 2, 1 \pmod 5$ respectively. Thus f has no roots over $\mathbb{F}_5$, and so clearly has no roots in $\mathbb{Z}_5$.

   (c) We first check that $f(X)$ has two roots (both 0 and 1) in $\mathbb{F}_2$. However we have $f'(0) \neq 0$, but $f'(1) = 0$, so Hensel's lemma cannot guarantee that $1 \in \mathbb{F}_2$ will lift to a root in $\mathbb{Z}_2$. So we must check $f(X)$ modulo 4. Now we observe that the only solution is $x = 0 \pmod 4$.

   Therefore, we have $v_2(f(0)) \geq 2 > 0 = 2v_2(f'(0))$. Thus Hensel's lemma tells us $f(X)$ has a unique solution in $\mathbb{Z}_2$.

2. Let $f(x) = (x^2-2)(x^2-17)(x^2-34)$. Clearly $f(x)$ has no root in $\mathbb{Q}$ and has a root in $\mathbb{R}$ (e.g. $\sqrt{2} \in \mathbb{R}$). We show that $f(x)$ has a root in $\mathbb{Q}_p$ for every prime p:

   - Case $p = 2$: Let $g(X) := X^2 - 17$. We note that $v_2(g(1)) = 4$ and $v_2(g'(1)) = 1$. Thus $v_2(g(1)) > 2v_2(g'(1))$, and so by Hensel's lemma, $g(X)$ and thus $f(X)$, has a root over $\mathbb{Z}_2$.
   - Case $p = 17$: We note that $X^2 - 2$ has a simple root mod 17, thus by Hensel's lemma, has a root in $\mathbb{Z}_{17}$.

- Case p odd, $p \neq 17$: Note that at least one of 2, 17, 34 must be a square mod p, as we have $\left(\frac{2}{p}\right)\left(\frac{17}{p}\right) = \left(\frac{34}{p}\right)$ by multiplicatively of the Legendre symbol. In particular, there is some $d \in \{2, 17, 34\}$ such that $X^2 - d$ has a root over $\mathbb{F}_p$. Again, by Hensel's lemma, this implies $f(X)$ has a root over $\mathbb{Z}_p$.

3. To show that the homomorphism $1 + p\mathbb{Z}_p \to 1 + p\mathbb{Z}_p$ given by $x \mapsto x^n$ is a bijection, it suffices to show that, for every $x \in 1 + p\mathbb{Z}_p$, there exists a unique $y \in 1 + p\mathbb{Z}_p$ such that $x = y^n$.

   Fix some $x \in 1 + p\mathbb{Z}_p$, and define the polynomial $f(X) := X^n - x$. Note that $f'(X) = nX^{n-1}$. Note that $f(1) \equiv 0 \pmod{\text{p}}$, and $f'(1) = n$ which is coprime to p. Thus we can apply Hensel's lemma which says there is a unique root y of $f(X)$ in $\mathbb{Z}_p$ such that $y \equiv 1 \pmod{\text{p}}$. This proves the claim.

4. (a) Let p be an odd prime. It's clear that if $\alpha \in (\mathbb{Z}_p^\times)^p$, then $\alpha \bmod p^2\mathbb{Z}_p \in ((\mathbb{Z}/p^2\mathbb{Z})^\times)^p$. Now assume $\alpha \in \mathbb{Z}_p^\times$ such that $\alpha \bmod p^2\mathbb{Z}_p$ lies in $((\mathbb{Z}/p^2\mathbb{Z})^\times)^p$.

   Thus, there exists some $y \in (\mathbb{Z}/p^2\mathbb{Z})^\times$ such that $y^p = \alpha \pmod{p^2}$. Let $f(X) := X^p - \alpha$. We note that $f'(X) = pX^{p-1}$. Now in order to apply Hensel's lemma, we need that $v_p(f(y)) > 2v_p(f'(y))$. As $v_p(f'(y)) = 1$, we thus need to show that $v_p(f(y)) \geq 3$. This will require a bit more work.

   Note that we can choose some $\beta \in \mathbb{Z}_p$ such that $\alpha \equiv y^p + p^2\beta \pmod{p^3}$. Now as y is a unit, we can let $\gamma := \beta/y^{p-1} \in \mathbb{Z}_p$.

   We now consider the following binomial expansion:

$$(y + p\gamma)^p = y^p + \binom{p}{1} y^{p-1} p\gamma + \binom{p}{2} y^{p-2} (p\gamma)^2 + \cdots + (p\gamma)^p$$

   Thus, taking the expansion modulo p^3, we obtain

$$\begin{aligned} (y + p\gamma)^p &\equiv y^p + p^2 y^{p-1}\gamma && \pmod{p^3} \\ &\equiv \alpha && \pmod{p^3} \end{aligned}$$

   noting that p is odd. This proves that α is in fact a p-th power modulo p^3, and then a standard application of Hensel's lemma, now using $X = y + p\gamma$, finishes the result.

   (b) Now let $p = 2$. Here we just do a standard application of Hensel's lemma. Let $\alpha \in \mathbb{Z}_2^\times$ such that $\alpha \bmod 8\mathbb{Z}_2$ lies in $((\mathbb{Z}/8\mathbb{Z})^\times)^2$. Thus there exists some $y \in (\mathbb{Z}/8\mathbb{Z})^\times$ such that $y^2 \equiv \alpha \pmod 8$. As before, we let $f(X) := X^2 - \alpha$. Here we have that $v_2(f(y)) \geq 3 > 2 = 2v_2(f'(y))$. Therefore, by Hensel's lemma, we have that $\alpha \in (\mathbb{Z}_2^\times)^2$.

   It's also worth noting that if $\alpha \bmod p^2\mathbb{Z}_p$ lies in $((\mathbb{Z}/2\mathbb{Z})^\times)^2$, then it's *not* guaranteed that $\alpha \in (\mathbb{Z}_2^\times)^2$. For example, 5 is a square modulo 4, but is *not* a square modulo 8.

5. Let $M = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ for some distinct primes $p_1, \ldots, p_k$. Now, using that the ideals $(p_1^{ia_1}), (p_2^{ia_2}), \ldots, (p_k^{ia_k})$ are pairwise coprime for any $i \geq 1$, then the

Chinese remainder theorem implies that

$$\mathbb{Z}/M^i\mathbb{Z} \cong \prod_{j=1}^{k} \mathbb{Z}/p_j^{ia_j}\mathbb{Z}$$

for every $i \geq 1$. Since inverse limits respect products, we can take inverse limits to obtain

$$\varprojlim_i \mathbb{Z}/M^i\mathbb{Z} \cong \prod_{j=1}^{k} \varprojlim_i \mathbb{Z}/p_j^{ia_j}\mathbb{Z}.$$

It thus suffices to show that $\varprojlim_i \mathbb{Z}/p_j^{ia_j}\mathbb{Z} \cong \mathbb{Z}_{p_j}$. One way to do this is simply by explicitly constructing the isomorphism maps:

$$\Psi_1 : \varprojlim_i \mathbb{Z}/p_j^{i}\mathbb{Z} \to \varprojlim_i \mathbb{Z}/p_j^{ia_j}\mathbb{Z}, \quad \text{given by} \quad (a_i)_{i\geq 1} \mapsto (a_{ia_j})_{i\geq 1}.$$

for each $j = 1, \ldots k$. Conversely, we can also construct the reverse maps:

$$\Psi_2 : \varprojlim_i \mathbb{Z}/p_j^{ia_j}\mathbb{Z} \to \varprojlim_i \mathbb{Z}/p_j^{i}\mathbb{Z}, \quad \text{given by} \quad (a_i)_{i\geq 1} \mapsto (a_{\lceil i/a_j \rceil} \bmod p_j^i)_{i\geq 1}.$$

for each $j = 1, \ldots k$. One can show that the two maps are bijections and that $\Psi_1 \circ \Psi_2 = \Psi_2 \circ \Psi_1 = \mathrm{id}$ from the compatibility conditions.

This therefore gives $\varprojlim_i \mathbb{Z}/M^i\mathbb{Z} \cong \prod_{j=1}^{k} \mathbb{Z}_{p_j}$, which proves the claim.[1]

6. (a) We can define a map $\mathbb{Z}_p \to [0,1]$ as

$$\sum_{i=0}^{\infty} a_i p^i \mapsto \sum_{i=0}^{\infty} a_i p^{-(i+1)}$$

We note that it is continuous, as if $x \equiv y \bmod p^n$, then $|f(x) - f(y)| \leq p^{-n}$. We also note it is surjective as any real $t \in [0,1]$ has a base-p expansion $t = \sum_{i=0}^{\infty} b_i p^{-i+1}$, where $b_i \in \{0, \ldots, p-1\}$.

(b) If there were to exist a continuous surjection $[0,1] \to \mathbb{Z}_p$, then this would imply $\mathbb{Z}_p$ is connected, as $[0,1]$ is connected.

But, we have that $\mathbb{Z}_p = \mathbb{Z}_p^\times \sqcup p\mathbb{Z}_p$, where $\mathbb{Z}_p^\times = \{x \in \mathbb{Z}_p \mid d(0,x) > 1/2\}$ and $p\mathbb{Z}_p = \{x \in \mathbb{Z}_p \mid d(0,x) < 1\}$, which implies $\mathbb{Z}_p$ is the disjoint union of two open sets (equivalently, the disjoint union of two closed sets), and is thus not connected.

One can more generally show that $\mathbb{Z}_p$ is in fact totally disconnected, i.e. the only connected subsets of $\mathbb{Z}_p$ are the singletons.[2]

7. Let us first assume that the sequence (a_n) is eventually periodic. In particular, we assume that (a_n) has (eventual) period k and that $a_{i+k} = a_i$ for all $i \geq M$.

[1] In general, one can show that the inverse limit $\varprojlim_i A_i$ over any inverse system $\{A_i\}$ is isomorphic to the inverse limit $\varprojlim_i B_i$ over any *cofinal* subsystem $\{B_i\}$ of $\{A_i\}$.

[2] In fact, $\mathbb{Z}_p$ is homeomorphic to the traditional Cantor set, for any prime p.

Therefore, we have

$$x = \sum_{n=N}^{\infty} a_n p^n = \sum_{n=N}^{M-1} a_n p^n + \sum_{n=M}^{\infty} a_n p^n = \sum_{n=N}^{M-1} a_n p^n + \sum_{j=0}^{\infty} \sum_{n=M}^{M+k-1} a_{n+jk} p^{n+jk}$$
$$= \sum_{n=N}^{M-1} a_n p^n + \left(\sum_{n=M}^{M+k-1} a_n p^n \right) \left(\sum_{j=0}^{\infty} p^{jk} \right).$$

noting that $a_{n+jk} = a_n$ for all $n \geq M$. However, as $\sum_{j=0}^{\infty} p^{jk} = 1/(1-p^k) \in \mathbb{Q}$, this implies $x \in \mathbb{Q}$.

Now conversely, assume that $x \in \mathbb{Q}$. Thus we can write $x = a/b$ for some coprime integers a, b. We can write $b = p^s m$ where m is coprime to p. We also choose $k \geq 1$ such that $p^k \equiv 1 \bmod m$ (e.g. we can pick $k = \mathrm{ord}_m(p)$). Our aim is to write x in the form

$$x = \frac{A}{p^s} + \frac{B}{1-p^k}$$

for some positive integers A and B. By multiplying out, we get the linear Diophantine equation

$$A(1-p^k) + Bp^s = a(1-p^k)/m \in \mathbb{Z}$$

Since $\gcd(p^s, 1-p^k) = 1$, we know there exists a solution in integers $A, B \in \mathbb{Z}$. In particular, as p^s and $(1-p^k)$ have opposite sign, we may assume A, B are positive integers.

Now let $A = \sum_{i=0}^{N} a_i p^i$ and $B = \sum_{i=0}^{M} b_i p^i$ be the (finite terminating) p-adic expansions of A and B respectively. Therefore, we have

$$x = \frac{A}{p^s} + \frac{B}{1-p^k} = \sum_{i=0}^{N} a_i p^{i-s} + \left(\sum_{i=0}^{M} b_i p^i \right) \left(\sum_{j=0}^{\infty} p^{jk} \right)$$

and so x has eventually periodic p-adic expansion.

8. We can show that $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2 \cong (\mathbb{Z}/2\mathbb{Z})^3$. To do this, recall from lectures that we have an isomorphism

$$\phi : \mathbb{Q}_2^\times \longrightarrow \mathbb{Z} \times (1 + 2\mathbb{Z}_2)$$
$$2^n u \longmapsto (n, u)$$

Now recall that Hensel's lemma tells us that squares in $\mathbb{Z}_2^\times$ are exactly $1+8\mathbb{Z}_2$. Thus the image of $(\mathbb{Q}_2^\times)^2$ under the map ϕ is $2\mathbb{Z} \times (1 + 8\mathbb{Z}_2)$. We therefore have that

$$\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2 \cong \mathbb{Z}/2\mathbb{Z} \times (1+2\mathbb{Z}_2)/(1+8\mathbb{Z}_2) \cong (\mathbb{Z}/2\mathbb{Z})^3.$$

We use the fact that, for any field K of characteristic $\neq 2$, all the quadratic extensions of L/K can be given by $L = K(\sqrt{d})$ for some non-trivial $d \in K^\times/(K^\times)^2$.[3] Thus, the quadratic extensions of $\mathbb{Q}_2$ are exactly:

$$\mathbb{Q}_2(\sqrt{-1}),\ \mathbb{Q}_2(\sqrt{2}),\ \mathbb{Q}_2(\sqrt{-2}),\ \mathbb{Q}_2(\sqrt{5}),\ \mathbb{Q}_2(\sqrt{-5}),\ \mathbb{Q}_2(\sqrt{10}),\ \mathbb{Q}_2(\sqrt{-10}).$$

[3] One could also apply Kummer theory in the case of $n = 2$, i.e. as $\mu_2 = \{\pm 1\} \subseteq K$, the quadratic extensions of K correspond to subgroups of order 2 in $K^\times/(K^\times)^2$